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BILLY 🪖: Steering Large Language Models via Merging Persona Vectors for Creative Generation

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Abstract

Multi-LLM systems enhance the creativity of large language models by simulating human collective intelligence but suffer from significant drawbacks, such as high computational costs and inference latency. To address these limitations, we propose **BILLY** (**B**lend**I**ng persona vectors for **L**arge **L**anguage model creativit**Y**), a training-free framework that captures the benefits of multi-LLM collaboration, *i.e.*, inducing diverse perspectives and specialized expertise, within a single model. BILLY operates by extracting and blending multiple distinct persona vectors directly in the model's activation space. We steer the model's generation process with this merged vector while inference, enabling multi-perspective output without explicit multi-LLM communication. Our experiments across creativity-oriented benchmarks demonstrate that BILLY surpasses single model prompting and traditional multi-LLM approaches, while substantially reducing inference time and computational costs. Our analyses further reveal that distinct persona vectors can be blended to achieve both effective control over complementary aspects of generation and greater interpretability. Our project page and codes are available at Website and GitHub.

1 Introduction

Creativity is widely recognized as a cornerstone of human progress, driving innovation across domains and enabling major scientific breakthroughs (Feist, 1998; Simonton, 2004). Extending this perspective, recent research (Franceschelli and Musolesi, 2023; Lin et al., 2025; Lu et al., 2026) has explored the creativity of large language models (LLMs), viewing them as promising tools for applications such as story writing (Gómez-Rodríguez and Williams, 2023; Chen et al., 2024b), design ideation (Ding et al., 2023; Hou et al., 2024; Hung et al., 2025), and scientific discovery (Yang et al., 2024b; Si et al., 2025), thereby augmenting human problem-solving and imagination (Zhang et al., 2023; Lim and Perrault, 2024; Liu et al., 2024, 2025).

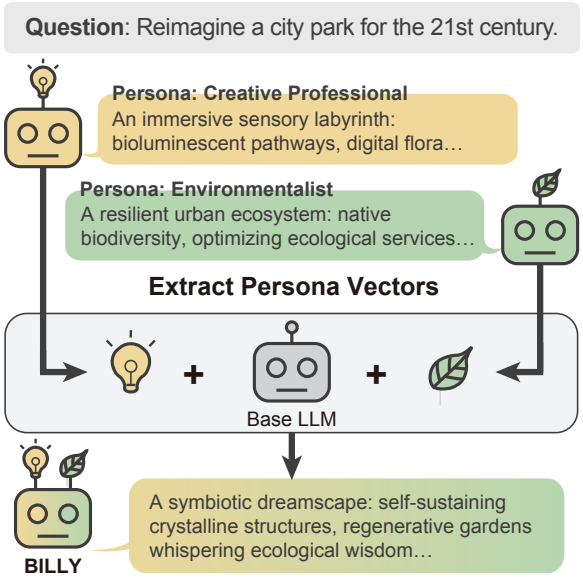

Figure 1: **BILLY** (**B**lend**I**ng persona vectors for **L**arge **L**anguage model creativit**Y**). To enhance the creativity of a single LLM, we extract and fuse the persona vectors of a `Creative Professional` and an `Environmentalist`, steering a base model by this composite vector to generate outputs based on both domains.

Recent studies highlight the potential of the *multi-LLM* paradigm (Guo et al., 2024; Lu et al., 2024; Lin et al., 2025), which aims to simulate human collective intelligence (Leimeister, 2010) by engaging multiple LLMs in iterative discussion to arrive at more comprehensive and well-balanced solutions (Tran et al., 2025). This paradigm allows systems to generate a broader range of ideas beyond a single model can reach (Tran et al., 2025). In the context of creativity, these frameworks often assign diverse roles to LLMs and employ structured, multi-round interactions (Lu et al., 2024; Lin et al., 2025; Su et al., 2025; Liang et al., 2024).

Our method, BILLY, takes its name from *The Minds of Billy Milligan* by Daniel Keyes—but the emoji in the title 🪖

For example, one LLM may act as a creative professional while another serves as an environmentalist, enabling the system to explore and integrate multiple perspectives. Through such iterative exchanges, the models collaboratively converge on creative solutions that balance innovation with sustainability.

Despite these advantages, multi-LLM frameworks also face a major challenge: the substantial computational cost and increased inference time associated with multi-LLM coordination (Tran et al., 2025). The iterative nature of multi-LLM dialogue substantially increases the system's input and output tokens, as models must consume and produce multiple rounds of text while processing other models' outputs. This multi-round exchange makes the process computationally costly and also increases inference latency (Lin et al., 2025; Creech et al., 2021; He et al., 2024), limiting scalability in real-world settings. Moreover, multi-LLM frameworks can exhibit sociological drawbacks such as *process loss* (Steiner, 1972), where the collective outcome falls short of the sum of individual contributions due to communication friction or coordination inefficiencies (Diehl and Stroebe, 1987; Nooteboom et al., 2007). Together, these issues expose a fundamental trade-off: while multi-LLM systems could potentially enhance diversity and creative potential, they do so at the expense of efficiency.

We therefore ask: *Without relying on multiple models carrying on a multi-round discussion, can a single LLM simulate diverse perspectives in an efficient manner?* A natural way to explore this idea is through prompting, that is, asking the model to simultaneously adopt different roles and viewpoints. For example, one might prompt an LLM to act as both a creative professional and an environmentalist when producing a solution that balances originality with sustainability. However, this approach depends heavily on the model's ability to integrate and coordinate multiple roles and perspectives within a single model and single response. In practice, an LLM may handle each persona separately but struggle to integrate them coherently in a single pass, often due to limitations in capability or the coverage of the training data (Turner et al., 2024).

To address these challenges while retaining the creative advantages of multi-LLM interaction, we propose **BILLY**(**B**lend**I**ng persona vectors for **L**arge **L**anguage model creativit**Y**) as illustrated in Figure 1. Inspired by recent advances in extracting *persona vectors* (Chen et al., 2025) and steering model behavior through *activation engineering* (Turner et al., 2024; Panickssery et al., 2024), BILLY extends these ideas by merging multiple persona vectors within a single LLM. This enables the model to produce multi-perspective, creative responses without resorting to computationally heavy multi-LLM frameworks. Notably, BILLY is training-free and empirically effective and efficient—it produces creative outputs that consistently reflect the targeted personas. Our experiments show that BILLY achieves higher creativity than single-model prompting or multi-LLM methods while being substantially more cost-efficient. We summarize our contributions as follows:

- **Enhanced Creativity.** We propose the framework BILLY, which blends multiple persona vectors within a single LLM. It enables the generation of diverse, multi-perspective, and creative responses, surpassing both single-model prompting and multi-LLM approaches on creativity-oriented benchmarks.
- **Efficiency and Simplicity.** BILLY is entirely *training-free*, requiring no additional fine-tuning or multi-LLM communication. It achieves comparable or superior creativity with substantially lower computational and token costs.
- **Interpretability via Persona Vectors.** Operating directly in the latent activation space, BILLY offers an interpretable mechanism for understanding and steering creativity, allowing fine-grained and transparent control through persona blending.

2 Related Work

2.1 LLM Creativity

Enhancing creativity in LLMs has been a growing research focus. One common approach involves role-playing prompts, which guide models to adopt specific creative personas (Chen et al., 2024a). Another works (Lu et al., 2024; Chen et al., 2024b; Su et al., 2025) explore multi-agent collaboration frameworks, where multiple agents simulate brainstorming sessions by proposing and critiquing ideas to achieve more sophisticated creative outcomes. Comprehensive reviews of this research direction can be found in Chakrabarty et al. (2023); Gómez-Rodríguez and Williams (2023).

nods to "Billie" Eilish instead.

2.2 Multi-LLM Collaboration

Multi-agent collaboration has emerged as a powerful paradigm for enhancing LLM applications (Du et al., 2023; Chan et al., 2024; Liu et al., 2023; Sun et al., 2023). AutoGen (Wu et al., 2023) enables next-generation applications through structured conversations, while LLM Discussion (Lu et al., 2024) and HOLLMWOOD (Chen et al., 2024b) employ role-play and dialogue to stimulate more creative outputs. Beyond creativity, researchers have explored emergent cognitive synergy through multi-persona self-collaboration (Wang et al., 2024) and applied divergent-agent systems to domain-specific tasks such as electronic design automation (Wu et al., 2025a). While infrastructures like AgentVerse (Chen et al., 2024c) and MetaGPT (Hong et al., 2024) provide frameworks for coordination, these collaborations can suffer from high computational costs, instability, and inefficiencies (Cemri et al., 2025; Han et al., 2025). The complexity of managing communication and coordination increases with the number of agents, potentially leading to bottlenecks and unpredictable behaviors.

2.3 Activation Steering

Activation steering is a technology that modifies a model's internal activations along specific directions to control or influence its behavior (Zou et al., 2025; Turner et al., 2024). This is achieved by first identifying a steering vector (Liang et al., 2026), which represents the desired behavioral shift. This vector is typically obtained using contrastive methods, where the model's activation differences are calculated from pairs of inputs that elicit opposite behaviors (e.g., a truthful vs. a deceptive response). During inference, this steering vector is added to the activations at a specific layer, effectively nudging the model's internal state toward the intended outcome.

A key advantage of this method is that it manipulates activations only at inference time without altering the model's underlying weights. This distinguishes it from techniques like model merging, which operates directly on model parameters to integrate capabilities from multiple fine-tuned models (Yang et al., 2024a; Sun et al., 2025; Jang et al., 2023), and offers a more lightweight, interpretable, and fine-grained control over model behavior. This technique has been applied across domains such as enhancing truthfulness (Li et al., 2023; Panickssery et al., 2024), improving safety (Zhang et al., 2025; Turner et al., 2024), and increasing transparency and interpretability (Zou et al., 2025; Jorgensen et al., 2023). Recent works have also leveraged activation steering for persona-related applications (Jorgensen et al., 2023; Turner et al., 2024; Zou et al., 2025). For example, Persona Vectors (Chen et al., 2025) identify activation directions corresponding to personalities for controllable character traits, while bi-directional preference optimization (Cao et al., 2025) enables versatile steering vectors for fine-grained persona control.

3 Approach: BILLY

Our approach, **BILLY**, **B**lend**I**ng persona vectors for **L**arge **L**anguage model creativit**Y**, is designed to overcome the limitations of prompting methods (Turner et al., 2024; Chen et al., 2024b; Liang et al., 2024; Hung et al., 2025) and the prohibitive costs of interactive multi-agent frameworks (Su et al., 2025; Ali et al., 2025; Lu et al., 2024). We introduce a novel paradigm that shifts the control of the model from an external, conversational system to direct, internal representational steering to improve LLM creativity. The core of our method is a three-stage process: (1) extracting distinct persona vectors through a contrastive method, (2) fusing these vectors offline into a single composite steering vector, and (3) applying this vector during inference to guide a single model.

3.1 Extracting Persona Vectors via Contrastive Activation

A *persona vector* (Chen et al., 2025) represents the characteristics of a specific persona (e.g., evil, humorous) as a directional vector within the model's activation space (Zou et al., 2025; Wu et al., 2025b; Panickssery et al., 2024), capturing the *shift* in the model's activation states when it adopts a persona compared to when it responds neutrally. By adding such a vector during inference, we can steer the model's behavior to align with the desired persona.

Building upon the pipeline proposed by Chen et al. (2025) for extracting persona vectors, we create a customized dataset designed to capture nuanced and specialized personas. This dataset enables more effective handling of creativity-oriented tasks and ensures a fair comparison with multi-LLM systems (Lu et al., 2024; Chen et al., 2024b; Ali et al., 2025).

Given a persona P, we curate two distinct sets of model responses, **positive-expression set** D_P^+ and